

Quality Plan – Sprint 1

Factory Inventory & Production Management System (MVP)

1. Purpose and Scope

This Quality Plan defines how quality will be assured and measured during Sprint 1, covering sprint deliverables, quality metrics, reviews, testing, and sign-off steps.

2. Sprint Context

Sprint Number: 1

Sprint Goal: Kickoff + foundations: repo/env setup, core auth + RBAC scaffolding, inventory CRUD baseline, and initial quality & risk documentation.

3. Roles and Responsibilities

Role	Owner	Quality responsibilities (Sprint 1)
Start-up manager	Nisal	Ensure sprint scope + documentation exist; coordinate demo readiness.
Quality manager	H J M S M jayasinghe	Own this plan; run reviews, tests, and QA sign-off; manage Quality Log.
Risk manager	Heshan Shalinda Koralagamage	Perform security/reliability checks; confirm RBAC and deployment risks mitigation.

4. Definition of Done (DoD)

- Acceptance criteria met for the user story / task
- Code reviewed (at least 1 reviewer) and merged to main
- Lint passes (no blocking ESLint errors)
- Unit/smoke tests pass (server + client)
- No critical security issues introduced (RBAC enforced server-side)
- Documentation updated where relevant (setup notes / env keys)
- Demo-ready increment (basic flow works end-to-end)

5. Sprint 1 Deliverables (Quality Targets)

- Repo + workspace setup
- Environment templates (.env.example)
- Project plan & product outline
- Team code of conduct
- Quality plan (this document)
- Risk management plan + risk log (separate)
- Auth wiring (Firebase)
- RBAC middleware + role seeding
- Inventory base models + validation
- Inventory CRUD API
- Inventory UI screens
- Audit logging skeleton
- Test framework baseline

6. Quality Metrics

Metric	How measured	Target
Build success	Both server and client build without errors	100% pass for sprint increment
Test pass rate	Automated tests status	All tests green for merged code
Defect leakage	Defects found during sprint review/demo	0 Critical, ≤2 High
Code review coverage	Merged PRs with review	≥90%
Doc completeness	Setup steps reproducible from docs	New dev can run locally < 30 min

7. Quality Activities and Review Cadence

Frequency	Activity
Daily	Quick sanity checks on in-progress features; verify lint/test on PRs
Mid-sprint	Quality checkpoint: review DoD adherence, open defects, and test coverage
End of sprint	Sprint QA sign-off: run smoke flow and verify sprint goal demo

8. Testing Approach (Sprint 1)

- Smoke flow: login → inventory CRUD → audit entry visible.
- API checks: status codes, validation errors, RBAC forbidden responses.
- Regression checks before sprint review: build + lint + test commands pass.

9. Defect Management

All defects and QA findings are recorded in the Sprint 1 Quality Log. Severity levels: Low, Medium, High, Critical. Critical/High issues must be fixed (or explicitly accepted) before sprint sign-off.

10. Sign-off

Quality sign-off owner: H J M S M jayasinghe. Sprint 1 is considered 'Done' when DoD is satisfied and the sprint goal demo is successful.